Part 11

2.6

a) $\ddot{p} = k_1 V_d = \dfrac{k_t}{2 J_p m_p l_p} V_d$

b) $\ddot{e} = k_2 \tilde{V}_s = \dfrac{k_t}{J_e (m_1 l_1^2 + 2 m_p l_h)} \tilde{V}_s$

$$\frac{d}{dt}\begin{bmatrix} p \\ \dot{p} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p \\ \dot{p} \\ \dot{e} \end{bmatrix} + k_t \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2 J_p m_p l_p} \\ \frac{1}{J_e (m_1 l_1^2 + 2 m_p l_h)} & 0 \end{bmatrix} \begin{bmatrix} \tilde{V}_s \\ V_d \end{bmatrix}$$

C_2

C_1

$k_t =$

$J_p =$

$J_e =$

$m_1 =$

$l_1 =$

$m_p =$

$l_h =$

$$u = Fr - Kx \qquad r = \begin{bmatrix} R_c \\ i_c \end{bmatrix} \qquad x = \begin{bmatrix} P \\ \theta \\ i \end{bmatrix}$$

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$$\dot{x} = Ax + BFr - BKx \qquad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\dot{x} = (A - BK)x + BFr$$

$$0 = (A - BK)x + BFr$$

$$0 = (\overbrace{AC}^{=0} - BKC + BF)r$$

$$0 = B(F - KC)r$$

$$F = KC$$

$$\begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} k_{11} & k_{13} \\ k_{21} & k_{23} \end{bmatrix}$$

Problem 4

$$Q = \begin{bmatrix} q_p & 0 & \\ 0 & q_{\dot{p}} & \\ & & q_\theta \end{bmatrix} \qquad R = \begin{bmatrix} r_{\tilde{v}_s} & 0 \\ 0 & r_{v_d} \end{bmatrix}$$

q

dim	Q	Max	Bryson Rule
p	q_p	1/6 rad	36
$\dot{p}$	$q_{\dot{p}}$	$\frac{3}{60}$ rad/s	400
θ	q_θ	$\frac{10}{60}$ rad	36

$$u = -Kx$$

$$x = \begin{bmatrix} p \\ \dot{p} \\ \theta \end{bmatrix}$$

$$u = \begin{bmatrix} \tilde{v}_s \\ v_d \end{bmatrix}$$

$$K = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \end{bmatrix}$$

Problem 5

$$\dot{x} = Ax + Bu + Gr$$
$$u = Fr - Kx$$

$$\frac{d}{dt}\begin{bmatrix} p \\ \dot{p} \\ \dot{\theta} \\ \gamma \\ \zeta \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & \gamma \\ 0 & 0 & -1 & 0 & 0 \end{bmatrix}}_{A} \begin{bmatrix} p \\ \dot{p} \\ \dot{\theta} \\ \gamma \\ \zeta \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{27_p m_p l_p} \\ \frac{1}{7_\theta (m_1 l_1^2 + 2 m_p l_1)} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_{B} \begin{bmatrix} \tilde{V}_\zeta \\ V_d \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}_{G} \begin{bmatrix} p_c \\ \dot{\theta}_0 \end{bmatrix}$$

$$y = Tx$$

$$\dot{x} = Ax + BFv - BKx + bv$$

$$0 = (A - BK)x + BFv + bv$$

$$0 = AIv - BKIv + BFv + bv$$

$$0 = B(F - KI) + b$$

$$F - KI = -B^{-1}b$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1/L_2 \\ 1/L_3 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 1/L_2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad A^2B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 & 0 & 0 & 1/L_2 & 0 & 0 \\ 0 & 1/L_2 & 0 & 0 & 0 & 0 \\ 1/L_3 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$